

Foundations of Cybersecurity

Part 1 of 3

- 1 What cybersecurity is, and why organizations need it
- 2 The three security properties
- 3 The lab environment



Thursday, September 17, 2026
Online class, 6:00 to 10:00 p.m.
Teacher: Asen Grozdanov





Before we start: use this material safely

1 Education only

Everything today is taught for defensive ends: to recognize, detect and protect. Nothing here authorizes anyone to touch a system without the permission of its owner.

2 Authorization first

Use the tools and techniques only on systems that you own, or that the owner has authorized you, **in writing**, to work on. Not certain? Then you do not have it.

3 Your responsibility

You alone answer for your actions, your equipment and your data. Work in the isolated lab, with fictional data only, and back up your files first.

Unauthorized activity may be a criminal offence: Criminal Code of Canada, sections 342.1, 342.2, 430(1.1) and 184.

Full terms: [Cyber.SoHo Legal Notice, Terms of Use and Acceptable Use Policy](#) · rules for this course: section 7.2





Today's plan: three lessons in two hours

0:00	0:05	0:42	1:19	1:53
	1 What cybersecurity is	2 The three security properties	3 The lab environment	

1 What cybersecurity is, and why organizations need it

The definition, six words of risk, Canadian figures, and a first asset register.

2 The three security properties

Confidentiality, integrity and availability: what each one means, what breaks it, what monitoring observes.

3 The lab environment

Virtual machines, network modes, the four environments of the program, and the readiness of your own computer.

Competency EZ01, first criterion: correct identification of issues linked to confidentiality, integrity and availability of information.



One fictional office, three deliverables

Grey Pine Bookkeeping

- A fictional office of **nine people** in Montréal.
- It keeps the books and prepares the tax returns of about **200 clients**.
- The office, its staff, its files and its addresses are invented for this course.
- Every example of today uses this office.

1

Deliverable 1: the asset register

Lesson 1. What the office owns, and a first order of priority. **Reused:** Lab 1 starts from this list.

2

Deliverable 2: ratings and baseline

Lesson 2. A rating of each asset for the three properties, and the reference hash of one sensitive file. **Reused:** Lab 1 uses the ratings; Labs 2 and 3 use the baseline.

3

Deliverable 3: the lab readiness report

Lesson 3. The proof that your computer can run the virtual machines. **Reused:** Labs 1, 2 and 3 all run in that environment.



1

What cybersecurity is and why organizations need it

At the end of this lesson you can:

- define cybersecurity in one sentence and name what it protects;
- use six words correctly: asset, threat, vulnerability, risk, control and incident;
- explain, with Canadian figures, why an organization of any size needs cybersecurity;
- say where preventative monitoring fits among the security activities;
- build a simple asset register and rank the assets by priority.



1 What is cybersecurity?

Cybersecurity is the protection of computers, networks, programs and digital information against six **unauthorized** actions. It combines technologies, written procedures and the daily practices of people.

access

use

disclosure

disruption

modification

destruction

unauthorized ...

Three neighbouring expressions

- **Information security:** information in every form, paper and speech included.
- **IT (information technology) security:** the computer equipment and the software.
- **Cybersecurity:** what is digital and connected to a network. The scope of this course.

Where security work begins

One question comes first: **who is allowed to do what, on which system?**

Sources: [glossary of the Canadian Centre for Cyber Security](#) and [glossary of the National Institute of Standards and Technology](#).



1 The vocabulary of risk: five words in order



The client files (**asset**) could be encrypted by a criminal group (**threat**) because the file server no longer receives updates (**vulnerability**); the likelihood is medium and the impact very high (**risk**), so the office installs a supported version and tests its backups every month (**controls**).

1 Risk = likelihood x impact

impact	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
		1	2	3	4	5
		likelihood (or exposure)				

- Each estimate receives a number from **1 to 5**.
- **Risk = likelihood x impact**. The result goes from 1 to 25.
- A threat without a vulnerability, or a weakness on an asset without value, produces little risk.
- A score helps to decide. It is **not a measurement**: write the reason for every rating.

15 or more: high

8 to 14: medium

7 or less: low

No organization brings every risk to zero. Find the highest risks and treat them first: **priority**.



1 Event, alert, incident

Event

Any occurrence that can be observed in a system or on a network.

Example: A user signs in.

Example: A file is opened.

Example: A disk reaches 80 percent.

Thousands per day. Nearly all are normal.



Alert

A message produced by a tool when events match a rule that somebody configured.

Example: "Five failed sign-ins on one account in one minute."

Produced by a rule, not by a person.



Incident

Events that harm or threaten information or systems, or that break a security rule.

Example: Client files copied to an unknown address outside the office.
Established by analysis.

An alert is a question. After analysis, the technician decides whether it describes an incident or not.



1 Where preventative monitoring fits

Prevention

Stops an unwanted action before it happens.

Detection

Notices quickly that something abnormal is happening.

Response

Limits the damage, corrects the cause, restores normal operation.

Preventative monitoring

The ongoing observation of a network in order to **detect and report problems before they cause damage.**

Three signals that announce a problem before it stops the work:

A disk that is 90 percent full

A backup that failed last night

Thirty failed sign-ins on the account of the owner



1 Why organizations need cybersecurity (1/2)

1 The work depends on digital systems.

Nobody prepares a tax return when the file server does not answer.

2 Incidents are frequent, and recovery costs are rising.

The national figures for 2023 are on the next slide.

3 The organization holds information that belongs to other people.

In Québec, Law 25 makes every enterprise responsible for the personal information it holds.

4 Clients, insurers and partners ask for proof.

An organization that cannot show what it protects loses clients and contracts.

5 The threat is constant, and it does not choose by size.

Automated searches find a small office exactly as they find a large company.



1 Why organizations need cybersecurity (2/2): Canada, 2023

1 in 6

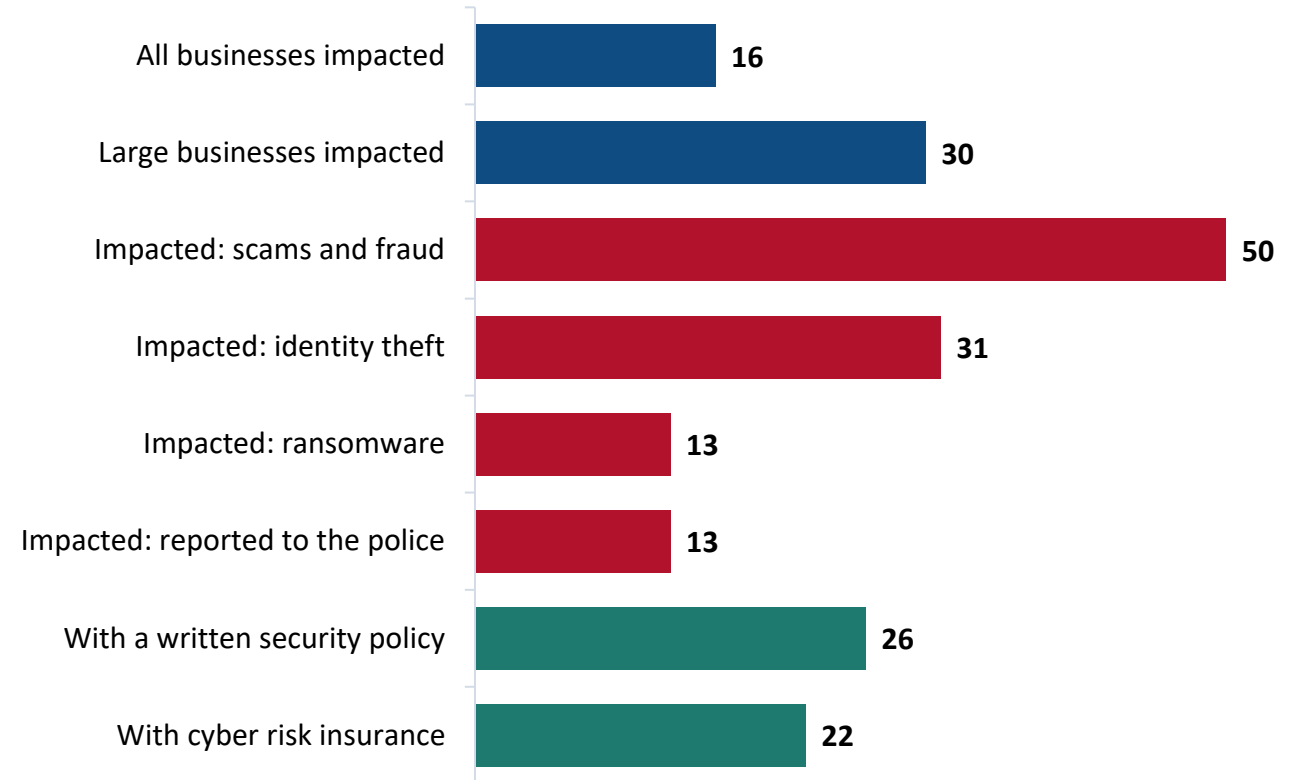
businesses impacted by cyber security incidents (16 percent)

1.2 billion \$

spent on recovery, double the amount of 2021

46 %

monitor their network and business systems: the most common activity



Percent of Canadian businesses, 2023. Source: [Statistics Canada, The Daily, October 21, 2024](#)



1 Who is responsible, and your role

Owner or management

Decides how much risk is acceptable, pays for the controls and answers for the result.

Administrator

Installs and maintains the systems. An employee or an outside provider.

Every employee

Applies the rules every day and reports what looks unusual.

Monitoring technician: you

Observes the network, detects problems early and reports them with evidence.

Your working conditions, from competency EZ01

- You work **under supervision**, with other technicians and with the users of the network.
- You are judged on professional ethics, management of priorities, clear communication, and **respect for the confidentiality of the monitoring data** that you collect.



1 Real-life example 1: the asset register

The owner asks: "What do we have that needs protection?"

And then: "If we can watch only a few things closely, which ones come first?"

1 List the assets

Walk through the office, ask each person, read the invoices. Eight assets: A1 to A8.



2 Rate the impact

From 1 to 5: what happens if the asset is lost, disclosed or altered?



3 Rate the exposure

From 1 to 5: how easily can a threat reach the asset?



4 Multiply and order

Score = impact x exposure.
High: 15 or more. Medium: 8 to 14. Low: 7 or less.

Result: Deliverable 1. Lab 1 (Friday, September 25) starts from this register to map the network and to decide what is monitored first.



1 Rating scales: impact and exposure

Rating	Impact: what happens if the asset is lost, disclosed or altered	Exposure: how easily a threat can reach the asset
1	A small inconvenience; the work continues.	Reachable by one administrator, from one computer.
2	Some tasks are slower for a few hours.	Reachable from one other system inside the office.
3	Part of the staff cannot work for a day.	Reachable by all the staff, inside the office only.
4	The whole office is slowed for several days, or restoring data becomes impossible.	Reachable from outside the office: Internet-facing equipment, radio signal, staff who browse the web and open attachments.
5	The office stops working, or personal information of clients is exposed.	Reachable by anyone on the Internet, at any time.

Exposure stands for likelihood in this first estimate.



1 Code example 1: the register in Python

1. It stores the **eight assets** with their two ratings.
2. It computes **score = impact x exposure** for each asset.
3. It assigns the band: high, medium or low.
4. It orders the list from the highest score to the lowest.
5. It prints the result and saves **greypine_asset_register.csv**.

Every line of code has a comment line before it. Run it only on your own computer or lab virtual machine.

Priority score = impact × exposure

Phase 5: the register is saved in a file

Rank	Asset	I × E	Score, from 0 to 25
1	A2 Cloud accounting application	5 × 4	20 HIGH
2	A3 E-mail accounts (9 users)	4 × 5	20 HIGH
3	A5 Router and firewall RTR01	4 × 4	16 HIGH
4	A1 File server FS01 - client files	5 × 3	15 HIGH
5	A4 Bookkeeper workstations (6)	3 × 4	12 MEDIUM
6	A6 Backup storage NAS01	4 × 2	8 MEDIUM
7	A7 Wi-Fi access point AP01	2 × 4	8 MEDIUM
8	A8 Network printer PRN01	1 × 3	3 LOW

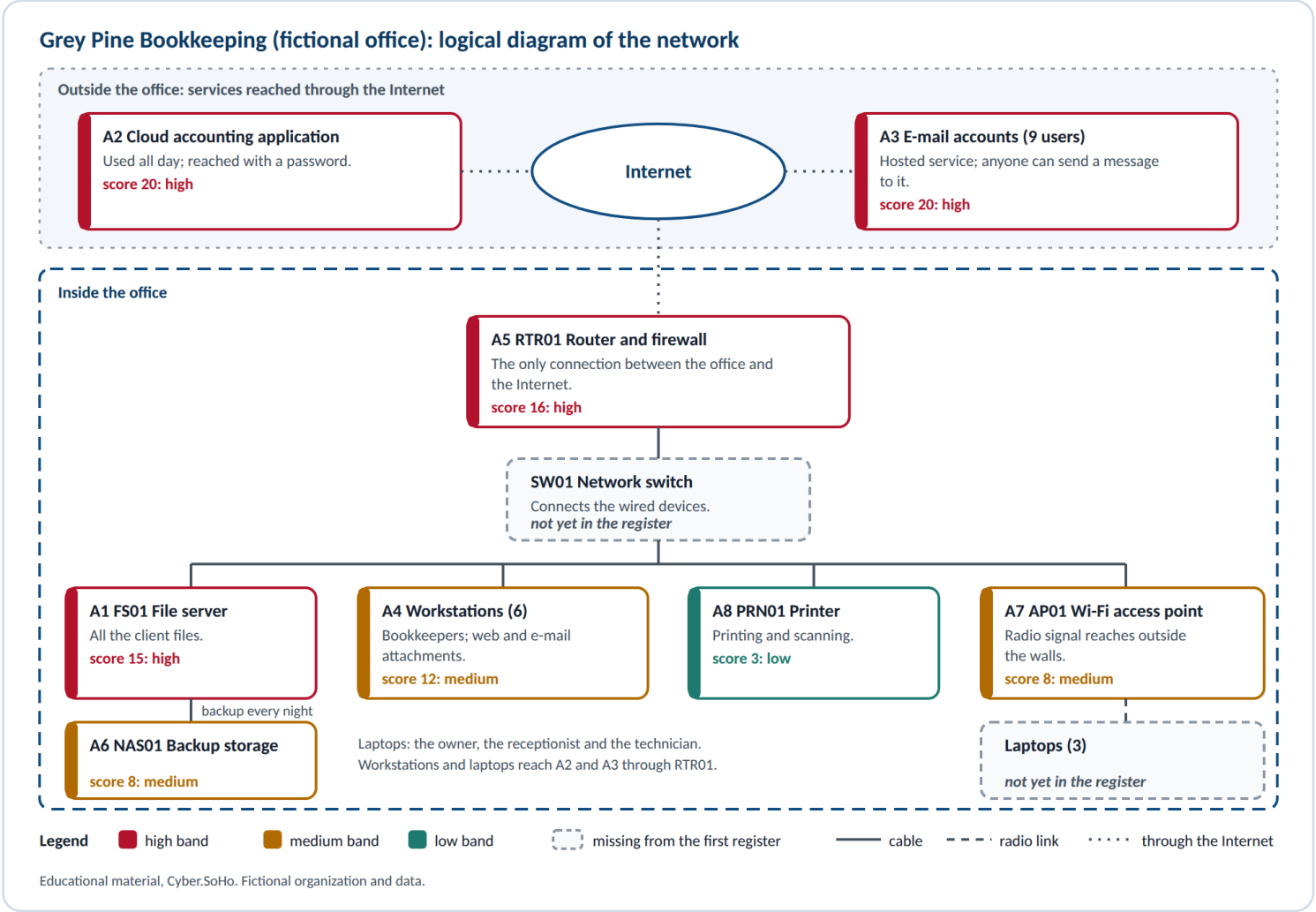
8: medium 15: high band starts

greypine_asset_register.csv: saved and closed, 9 rows



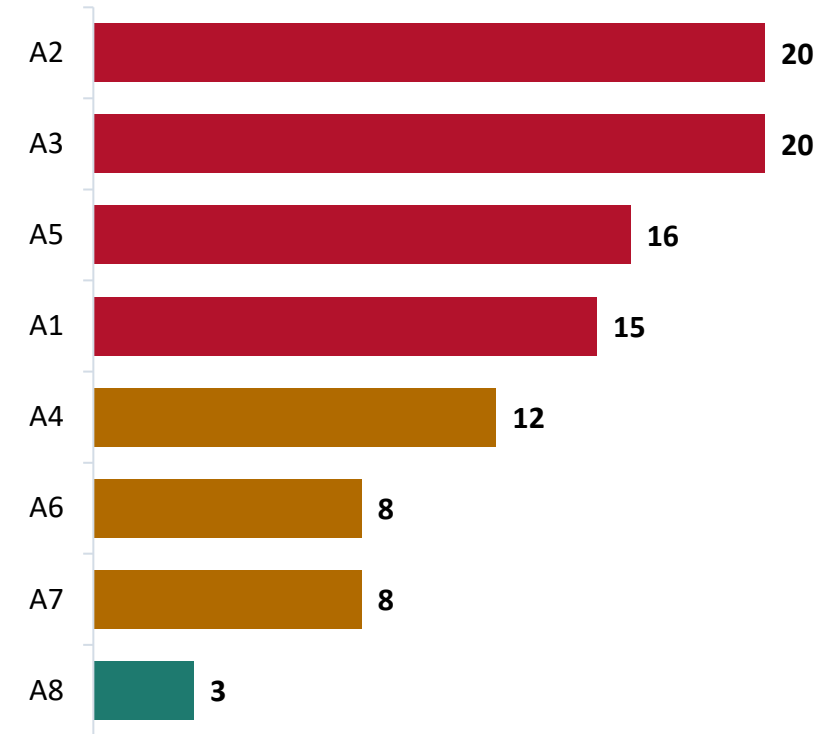
1

Figure 1: the office network of Grey Pine



1 Deliverable 1: the asset register

#	Id	Asset	I	E	Score	Band
1	A2	Cloud accounting application	5	4	20	high
2	A3	E-mail accounts (9 users)	4	5	20	high
3	A5	Router and firewall RTR01	4	4	16	high
4	A1	File server FS01, client files	5	3	15	high
5	A4	Bookkeeper workstations (6)	3	4	12	medium
6	A6	Backup storage NAS01	4	2	8	medium
7	A7	Wi-Fi access point AP01	2	4	8	medium
8	A8	Network printer PRN01	1	3	3	low



I = impact, E = exposure. The file server has the highest impact but ranks fourth: it cannot be reached from the Internet.



1 Lesson 1: video and practice

Watch after the class

- [Cyber Security in 7 Minutes](#), Simplilearn, about 7 minutes.
- It repeats the vocabulary of this lesson in a different voice and announces the class of Friday on the types of attacks.

Practice in the lecture page

- **5 flash cards:** answer aloud, then turn the card.
- **6 questions:** four with one answer, two with several; each has a hint and an explanation.
- Practice only. Quiz 1 is a separate document.

Try this one now

The whole staff of Grey Pine uses one password for the accounting application. In the vocabulary of risk, the fact that this password is shared is:

- A.** a threat
- B.** a vulnerability
- C.** an asset
- D.** a control



2

The three security properties: confidentiality, integrity, availability

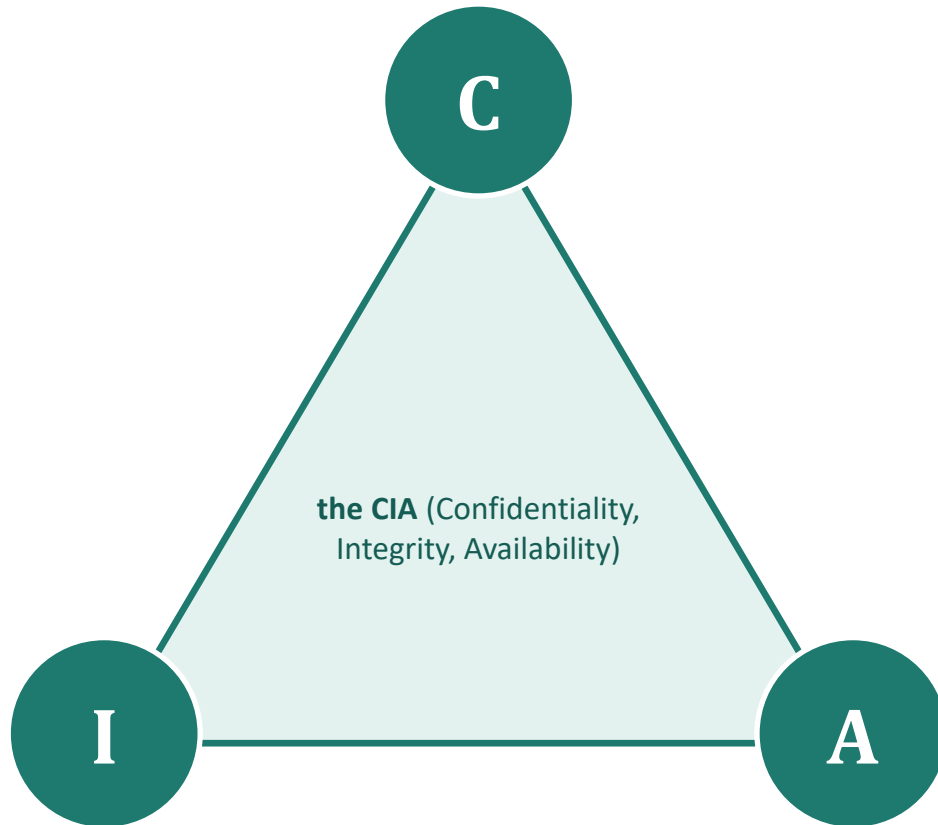
At the end of this lesson you can:

- define the three properties and recognize which one is lost in a given situation;
- name the usual controls that protect each property;
- explain what a hash is and what a comparison with a baseline proves;
- rate an asset as low, moderate or high for each property, and apply the high-water mark rule;
- say what a monitoring tool observes for each property.



2

The three security properties



Confidentiality

Who is able to read this? Losing it means unauthorized **disclosure**.

Integrity

Is this exact and complete, and was every change authorized? Losing it means unauthorized **modification or destruction**.

Availability

Can authorized people use this when they need it? Losing it means **disruption** of access or of use.

For any alert: was something **read, changed**, or made **unreachable**?



Confidentiality

Confidentiality: information is not made available or disclosed to persons, programs or organizations that are not authorized to have it.

What breaks it at Grey Pine

- A tax return is sent to the wrong client.
- A laptop without disk encryption is stolen.
- A former employee still has a working account.
- A criminal reads a mailbox for three weeks: nothing changes, nothing stops.

Controls that protect it

- **Authentication**, with **MFA** (multi-factor authentication).
- **Least privilege:** only the rights required for the work.
- **Encryption** at rest and in transit.
- Accounts closed on the day of departure.

What monitoring observes

- Failed and successful sign-ins.
- Access to sensitive folders.
- Changes of permissions.
- New accounts, above all administrator accounts.
- Large transfers toward the outside.



2 Integrity

Integrity: information is exact and complete, and information and systems are changed only by authorized persons, in authorized ways.

What breaks it at Grey Pine

- An amount is altered in an invoice.
- A payee is added to the payroll file without approval.
- Malicious software replaces a system file.
- An old version overwrites the current one by mistake.
- Lines of a log are erased to hide an action.

Controls that protect it

- Write permissions limited to those who need them.
- **Digital signatures.**
- Version history.
- **Change management:** requested, approved, recorded.
- Backups, to return to a correct version.

What monitoring observes

- Hashes of protected files compared with a **baseline**.
- Changes of configuration.
- Gaps or erasures in the logs.
- This is **FIM** (file integrity monitoring).



2

How a change is detected: the hash

A file of any size



Hash function SHA-256 (Secure Hash Algorithm, 256-bit)



64 hexadecimal characters: the hash

Four properties

- **Deterministic:** the same input always gives the same hash.
- **Fixed length:** one line or 4 gigabytes both give 64 characters.
- One changed character gives a **completely different** hash.
- **One direction only:** the input cannot be rebuilt from the hash.

The method of the baseline

Record the hash of the approved file: the **baseline**. Later, compute it again. **Equal:** not a single byte changed. **Different:** the content changed.

What it does not say

Not **who**, not **when**, not **what**, not whether the change was **approved**. A hash is not encryption: it hides nothing.



Availability

Availability: information and systems can be reached and used by authorized persons when they need them, within an acceptable time.

What breaks it at Grey Pine

- Ransomware encrypts the client files.
- A **DDoS** (distributed denial of service) attack saturates the Internet link.
- The disk of the file server is full.
- The power fails.
- An update in the tax season stops the accounting application.

Controls that protect it

- **Tested backups**, 3-2-1 rule.
- A **UPS** (uninterruptible power supply).
- Duplicate equipment for what cannot stop.
- Updates at a planned time, tested first.
- Enough disk, memory and network capacity.

What monitoring observes

- Whether each device and each service answers.
- Use of processor, memory, disk and link.
- The result of every backup.
- Remaining validity of certificates and licences.

2

How availability is measured

99 %

uptime allows an interruption of **87.6 hours per year, about 3.65 days**

99.9 %

uptime allows an interruption of **8.76 hours per year**

99.99 %

uptime allows an interruption of **about 53 minutes per year**

RTO (recovery time objective)

The longest time that the organization accepts to remain without a system after an interruption.

RPO (recovery point objective)

The largest amount of recent work, as a duration, that it accepts to lose. 24 hours requires one backup per day.

The 3-2-1 rule

3 copies of the data, on **2** different kinds of storage, with **1** copy outside the office. And the backups are tested.



2 Which property is lost?

Situation at Grey Pine	C	I	A
A bookkeeper sends a tax return to the wrong client.	lost		
A fourth payee is added to the payroll file without approval.		lost	
The Internet link is saturated for three hours.			lost
The disk of the file server is full; nobody can save a document.			lost
A laptop without disk encryption is stolen from a car.	lost		lost
Ransomware encrypts the client files, and the criminals first copy them.	lost	lost	lost

C = confidentiality, I = integrity, A = availability. No attacker is needed to lose a property: a full disk is enough.



2

Rating the impact: low, moderate, high

Low

The effect is limited. The work continues with a minor loss.

Moderate

The effect is serious: an important activity is degraded for a significant time.

High

The effect is severe or catastrophic: the main activity stops, or persons suffer serious harm.

The high-water mark rule

- The overall level of an asset is the **highest** of its three levels.
- **Router RTR01**: confidentiality low, integrity high, availability high. Overall: **high**.
- One serious weakness is enough to cause serious harm.
- Source: **FIPS** (Federal Information Processing Standards) Publication 199.

The properties compete

- Every extra proof of identity protects **confidentiality**, and slows or blocks a legitimate user: less **availability**.
- One password shared by the staff gives quick **availability**, and destroys **confidentiality**.
- There is no universal setting: each organization decides.



2 From the three properties to monitoring

Property	Question of the technician	What the tools observe	Example of an alert rule
Confidentiality	Did somebody read or receive information that he should not have?	Sign-ins that fail and succeed; access to sensitive folders; changes of permissions; new accounts; large transfers toward the outside.	More than five failed sign-ins on one account in one minute.
Integrity	What changed, and was the change approved?	Hashes of protected files compared with a baseline; changes of configuration; gaps or erasures in the logs.	The hash of a protected file differs from its baseline.
Availability	Can people reach what they need, now and tomorrow?	Answer of each device and service; use of processor, memory, disk and link; result of backups.	Disk used above 90 percent; last backup failed.

Each row becomes alert rules in **Lab 2** and material for analysis in **Lab 3**.



2 Real-life example 2: one payroll file, three checks

September 15: approval

- The owner approves **payroll_2026-09.csv**: three employees.
- Permissions set to **640**: owner and payroll group only.
- The hash is recorded: the **baseline**.

September 17: routine check

- Permissions are now **644**: every user of the server can read the file.
- The hash is **different**.
- A fourth row: an unknown payee, 9,999 dollars.

What the technician does

- He **records the evidence**: two hashes, two permission values, the time.
- He **informs the owner** immediately.
- He keeps the file **as it is** for the analysis.

Report the evidence. Do not repair in silence. A silent correction destroys the evidence and can hide a fraud.



2

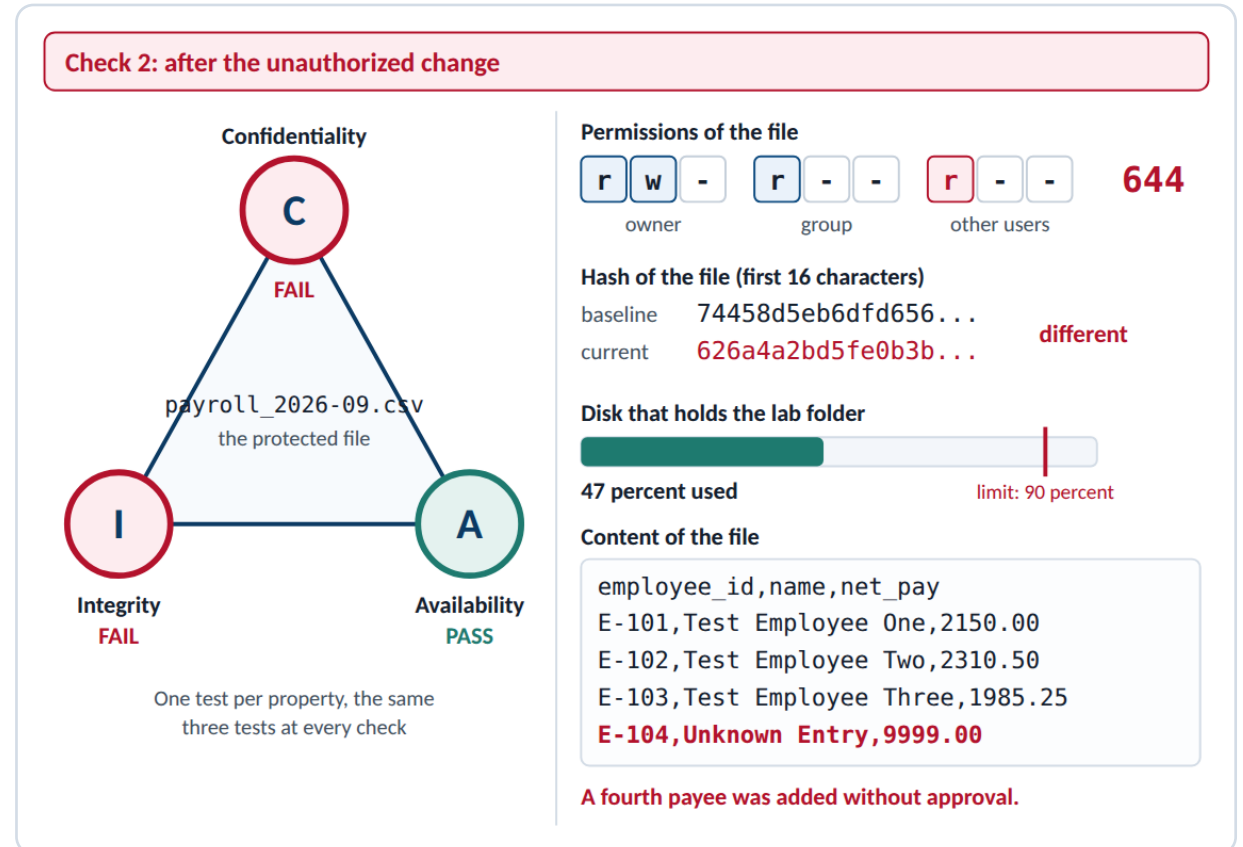
Code example 2: three checks in Bash

- It creates the fictional payroll file and sets permissions **640**.
- It records the baseline hash.
- One function runs **one test per property**.
- It simulates the unauthorized change and runs the three tests again.

```

--- Check 1: right after the baseline ---
C PASS permissions 640
I PASS hash equals the baseline
A PASS file is readable - disk 47% used
--- Check 2: after the unauthorized change ---
C FAIL permissions 644
I FAIL hash differs from the baseline
A PASS file is readable - disk 47% used

```



2 Figure 2: the three properties applied to one file

The three properties applied to one file			
payroll_2026-09.csv			
	Confidentiality Who is able to read the file?	Integrity Is the content exact, and was every change approved?	Availability Can the payroll group use it when needed?
What can go wrong	• Every user of the server can open the file. • A former employee still has an account. • The file is sent to the wrong person.	• A payee is added without approval. • An amount is altered. • An old version replaces the current one.	• The disk of the server is full. • Ransomware encrypts the file. • The server does not start after an update.
Controls that protect it	• Permissions 640: owner and payroll group only. • One account per person, with a second proof of identity. • Accounts closed on the day of departure.	• Write right limited to the payroll group. • Every change approved by the owner. • Version history and backups.	• Tested backups: 3 copies, 2 kinds of storage, 1 outside the office. • Battery unit for short power failures. • Updates at a planned time.
What monitoring watches	• Changes of permissions. • Access to the payroll folder. • Failed and successful sign-ins.	• Hash of the file compared with the baseline. • Date of the last change. • In the logs: who wrote to the file.	• Disk space used. • Answer of the server. • Result of last night's backup.
Test in the script	• Last digit of the permissions: 0 passes; 4 or more fails.	• Hash equal to the baseline passes; a different hash fails.	• File readable and disk under 90 percent: passes.

Educational material, Cyber.SoHo. Fictional organization and data.



2 Deliverable 2: ratings and baseline

Id	Asset	C	I	A	Overall
A1	File server FS01	high	high	moderate	high
A2	Cloud accounting application	high	high	high	high
A3	E-mail accounts	high	moderate	moderate	high
A4	Workstations (6)	moderate	moderate	moderate	moderate
A5	Router and firewall RTR01	low	high	high	high
A6	Backup storage NAS01	high	high	moderate	high
A7	Wi-Fi access point AP01	moderate	moderate	low	moderate
A8	Network printer PRN01	moderate	low	low	moderate

Baseline record

File: payroll_2026-09.csv

Permissions expected: 640

Hash function: SHA-256 (Secure Hash Algorithm, 256-bit)

Baseline, September 15: 74458d5eb6dfd656...

Observed, September 17: 626a4a2bd5fe0b3b...
different: **report**

C, I, A: the three properties. The backup storage had a medium score in Lesson 1 (low exposure); here it is **high**, because it holds a copy of everything. Lab 1 uses both views.



2

Lesson 2: videos and practice

Watch after the class

- [The three properties in five minutes](#), Professor Messer, 5 minutes 17 seconds. Watch this one first.
- [Fundamentals of the three properties](#), IBM (International Business Machines) Technology, 12 minutes 25 seconds.

Practice in the lecture page

- **5 flash cards:** answer aloud, then turn the card.
- **6 questions:** four with one answer, two with several; each has a hint and an explanation.
- Practice only. Quiz 1 is a separate document.

Try this one now

On September 17 the hash of the payroll file differs from the baseline of September 15. What does this prove?

- A.** The account of the owner was stolen.
- B.** The content changed after the baseline was recorded.
- C.** Ransomware encrypted the file.
- D.** An unauthorized person read the file.



3

The lab environment: VirtualBox, VMware Workstation Pro, the cloud lab and CyberQuébec

At the end of this lesson you can:

- explain why security work is practised in a separate lab, and state the three conditions of an Authorized Lab Environment;
- define virtualization, hypervisor, host, guest, snapshot and clone;
- choose the network mode of a lab virtual machine, and explain why the bridged mode is not used;
- describe the four environments of the program and the use of each one;
- check whether your own computer can run the labs, and produce the readiness report.



3 Why a separate lab

1 A legal reason

On a shared network, a capture records the communications of other people.
Legal Notice, sections 6.1 and 7.2.

2 A technical reason

An exercise can fill a disk or stop a service.
In a lab, the machine is rebuilt in a few minutes.

3 A teaching reason

The same machines, names and addresses for everyone: results can be compared, errors are found quickly.

Authorized Lab Environment: three conditions

- (a) You **own** it, or you are expressly permitted to use it for the exercise.
- (b) It is **separated** from every production, institutional, public or third-party network.
- (c) It contains **no real personal data**, and no system of a person who has not consented in writing.

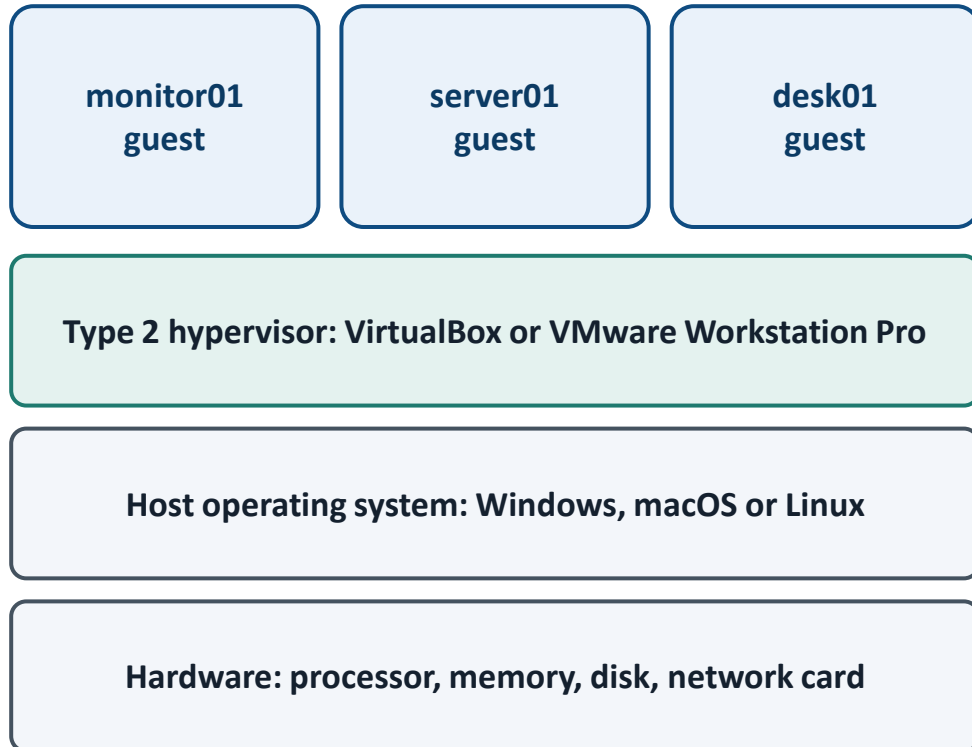
Designated targets

Work only on the machines, addresses and files that the lab guide names.
A machine does not become a target because it can be reached.



3

How virtualization works



- **Virtualization:** software divides the processor, memory, disk and network of one physical computer among several isolated environments.
- **Virtual machine:** one of these environments, with its own virtual hardware and its own operating system.
- **Hypervisor:** the software that creates and runs the virtual machines.
- **Type 1** is installed directly on the hardware: data centres, cloud providers.
- **Type 2** is an ordinary application on Windows, macOS or Linux: the type of this course.

The **host** is the physical computer. A **guest** is an operating system that runs inside a virtual machine.



3 Virtual disk, installation image, snapshot, clone

1 Virtual disk

A large file on the host. For the guest, it is a complete hard disk. Easy to copy, move and delete.

2 Installation image

A file, usually **.iso**, with an exact copy of an installation disc. Download it from the publisher only, then **verify its checksum**.

3 Snapshot

A recorded state of a virtual machine, to which you can return. Take one **before every exercise**. It is **not a backup**.

4 Clone

An independent copy of a virtual machine with its own name. Install one clean Linux machine, then clone it for each role.



3 What your computer needs

Resource	Minimum	Recommended	Why
Memory	8 gigabytes	16 gigabytes or more	With 8, two small virtual machines run together; with 16, the three machines of the standard layout.
Processor cores	4	6 or more	The host keeps two cores for itself and lends the others.
Hardware virtualization	enabled	enabled	Without it, 64-bit guests do not start.
Free disk space	80 gigabytes	100 or more, on a solid-state drive	Three virtual disks, their snapshots and the installation images.
Hypervisor	one	one	VirtualBox or VMware Workstation Pro. One is enough.

Intel names the processor feature **VT-x** (Virtualization Technology for x86 processors); its competitor names it **AMD-V** (AMD Virtualization). It is switched on in the firmware settings, the **UEFI** (Unified Extensible Firmware Interface). Guide: [Enable virtualization on Windows](#).



3

Network modes of a virtual machine, and the rule of this course

NAT (Network Address Translation)

Reaches the Internet through the address of the host. Nobody reaches it from outside.

Only for a short time, to download updates.

Bridged

Joins the physical network: home, College or workplace. Every device there can reach it.

NOT USED. Lab traffic would mix with the traffic of other people.

Host-only

Reaches the host and the other machines of the same private network. No Internet.

THE STANDARD MODE for all lab traffic.

Internal network

Reaches only the virtual machines attached to it. Not even the host.

For exercises that need the strictest isolation.

The rule

Lab traffic travels on a **host-only** or **internal** network. A network address translation adapter is connected only for an update. In **promiscuous mode** a card accepts every frame: on a shared network this records other people (Criminal Code, section 184).



3

The four environments of the program

Oracle VirtualBox

- Type 2 hypervisor, free and open source.
- Latest release on September 17, 2026: **7.2.18**.
- The Extension Pack is **not needed**.
- **Default choice** of the course.

VMware Workstation Pro

- Type 2 hypervisor, free since November 2024; no licence key.
- Latest release: **26H1u1**.
- Download with a free Broadcom account. **Register today**.

Learner Lab of AWS (Amazon Web Services) Academy

- A temporary cloud account, by invitation of the instructor.
- Sessions of **4 hours**; a **budget** of credits; two regions.
- Replaces the local lab when needed.

CyberQuébec range

- A practice environment that reproduces the network of an organization.
- **Not available yet** for this cohort.
- The instructor will give the account and the rules.



3

The rules of a safe lab

1. **Isolate.** Virtual machines, host-only or internal network.
2. **Snapshot and back up** before every exercise.
3. **Use only the designated targets** that the lab guide names.
4. **Keep it fictional.** No real person, employer or password.
5. **Official sources and valid licences** only; verify the checksum.
6. **Respect College equipment and cloud terms.**
7. **Clean up:** stop the tools, delete what you do not need.

The most likely mistake

Running a capture on the home or campus Wi-Fi "to have real traffic to analyze".

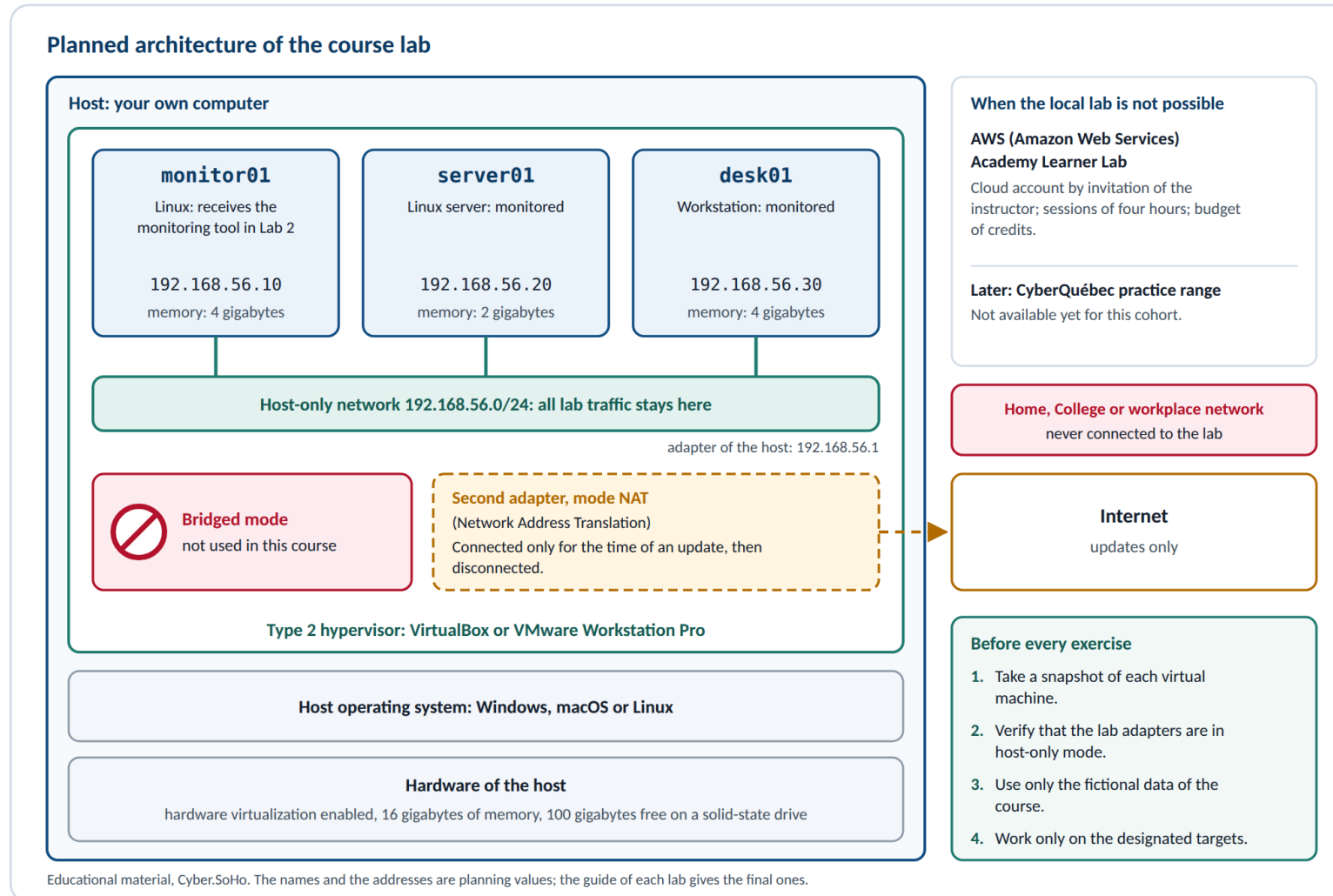
That traffic belongs to other people.

The labs provide all the traffic that you need.

Before the first lab you sign the student acknowledgement, **Annex A** of the [Legal Notice](#).



3 Figure 3: planned architecture of the course lab



3 Real-life example 3: is this computer ready?

Check	Required
Installed memory	8 gigabytes or more
Processor cores	4 or more
Hardware virtualization	enabled
Free space on drive C:	80 gigabytes or more

Hypervisor installed at least one

Computer A

16 gigabytes, 8 cores, 212 free, VirtualBox.

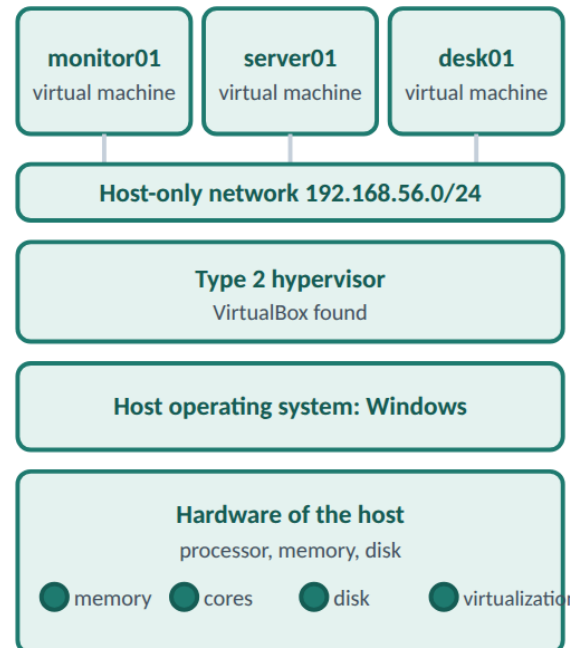
READY

Computer B

Virtualization off, 61 free, no hypervisor.

NOT READY

Computer A: LAPTOP-GP07



Windows hypervisor active: False
Information only: nothing to do.

The five checks

Installed memory PASS
measured: 16 gigabytes
required: 8 gigabytes or more

Processor cores PASS
measured: 8 cores
required: 4 or more

Hardware virtualization PASS
measured: enabled = True
required: enabled = True

Free space on drive C: PASS
measured: 212.4 gigabytes
required: 80 gigabytes or more

Hypervisor installed PASS
measured: VirtualBox = True, VMware = False
required: at least one

Verdict: READY

lab_readiness_report.csv: saved, 5 rows



3 Deliverable 3, and what to do if a check fails

Failed check	Usual cause	What to do
Installed memory	Less than 8 gigabytes.	Add memory if the model allows it. Otherwise tell the instructor and use the cloud lab.
Processor cores	An older processor with two cores.	Run one virtual machine at a time, or use the cloud lab.
Hardware virtualization	Switched off in the firmware settings.	Enable it, then run the check again.
Free space on drive C:	The disk is nearly full.	Free some space, or store the virtual machines on a second drive.
Hypervisor installed	Nothing installed yet.	Install VirtualBox or VMware Workstation Pro from the official site.

Keep the file **lab_readiness_report.csv**: it is Deliverable 3. Verdict NOT READY? Write to the instructor **before Lab 1**.



3 Set up your lab this week



Official pages

- [VirtualBox: download page](#)
- [Broadcom: download steps for VMware Workstation Pro](#)
- [Learner Lab portal](#)
- [Microsoft: Get-FileHash](#)

Companion page

Lab environment setup, step by step: nine animations that show every step, with a note on versions under each one.

Versions checked on September 17, 2026: VirtualBox **7.2.18**, VMware Workstation Pro **26H1u1**.



3

Lesson 3: videos and practice

Watch after the class

- [Virtualization Explained](#), IBM (International Business Machines) Technology, 5 minutes 19 seconds.
- [How to Install VirtualBox on Windows 11](#), GEEKrar, about 2 minutes.

Practice in the lecture page

- **5 flash cards:** answer aloud, then turn the card.
- **6 questions:** four with one answer, two with several; each has a hint and an explanation.
- Practice only. Quiz 1 is a separate document.

Try this one now

You prepare monitor01 and server01 for an exercise that captures their traffic. Which network mode do you choose?

- A. Bridged
- B. Host-only
- C. Network address translation, because it is the default mode
- D. No network adapter at all





Lab 1 in summary

Friday, September 25

Class 7, online

5 percent

of the final grade

Part 1

of the Integrative Project

1. Read the documentation and draw the **logical diagram**.
2. Complete the **asset register**, with the items that a first inventory misses.
3. Rate each asset for the **three properties**; apply the high-water mark.
4. Note the **constraints**: hours, budget, staff, legal duties.
5. State **what must be monitored** for each important asset.
6. Put the monitoring activities in **order of priority**.
7. Write the **monitoring plan** for the owner.

A summary only

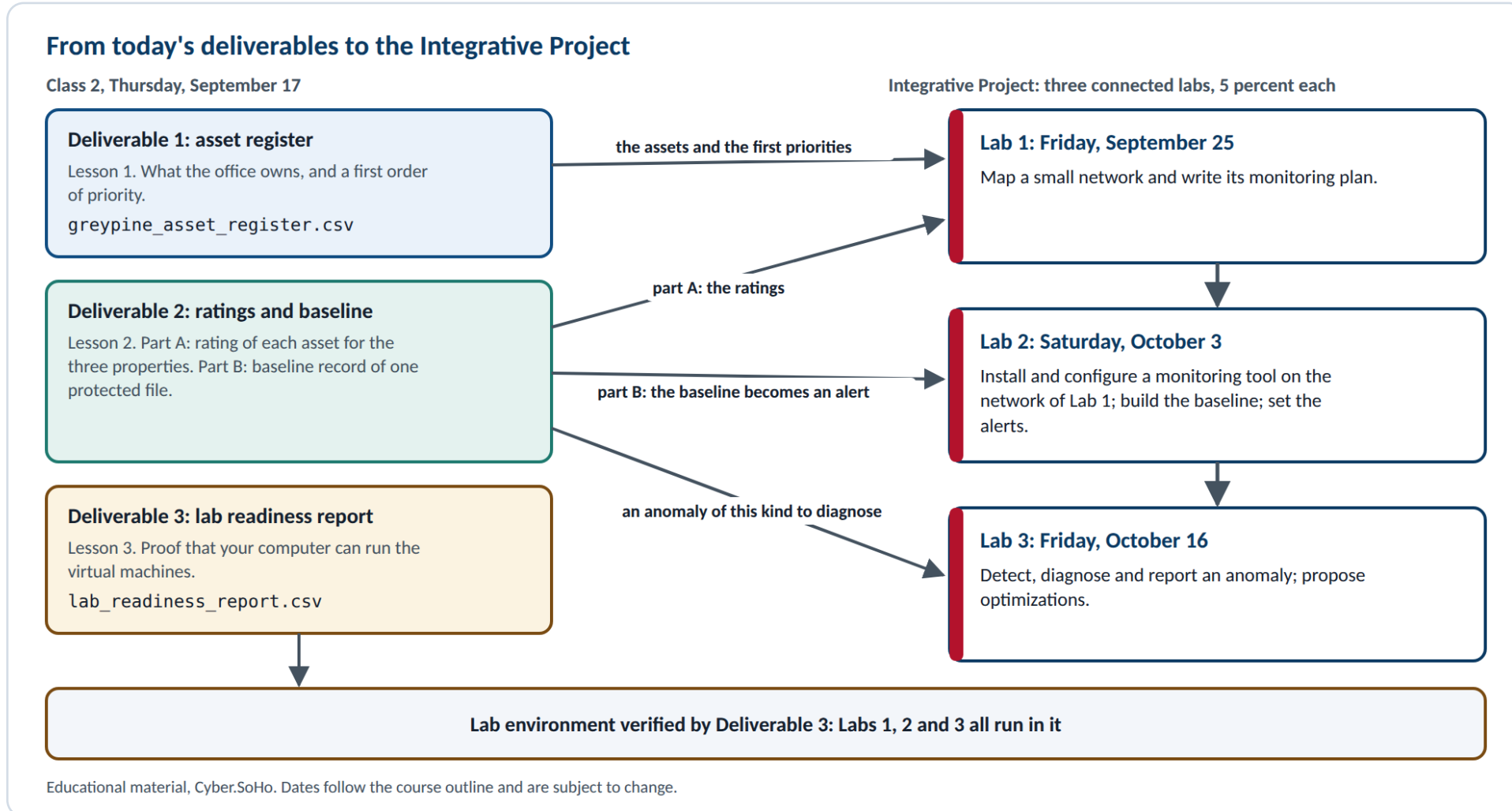
No tool is installed and no traffic is captured: that work belongs to Lab 2.

The official instructions of Lab 1 are published separately before September 25. Where they differ from this summary, **the official instructions apply**.





Figure 4: from today's deliverables to the three labs





Key points of today

1. Cybersecurity protects digital systems and information against **six unauthorized actions**.
2. A **risk** combines likelihood and impact. The highest risks are treated first.
3. An **event** is observed, an **alert** comes from a rule, an **incident** is established by analysis.
4. Organizations of every size need it: **1 in 6** Canadian businesses impacted in 2023.
5. Every alert leads to three questions: **read, changed, unreachable?**
6. A hash compared with a baseline proves **that** the content changed, not who, when or why. **Report, do not repair in silence.**
7. Each asset is rated low, moderate or high; the **highest** level is its overall level.
8. Lab work runs in virtual machines on a **host-only** network, with snapshots and fictional data.
9. Four environments: VirtualBox, VMware Workstation Pro, the cloud lab and, later, CyberQuébec.





Before the next class

Friday, September 18

Sub-topics 1.3 and 1.4: types of cyberattacks; prevention, detection, response.

Saturday, September 19

Sub-topic 1.5, end of Topic 1, and **Quiz 1 (5 percent)**.

1. Read the [Legal Notice](#), in particular sections 6, 7.2 and 9; sign Annex A when the instructor sends the page.
2. Install **one** hypervisor from its official site, and verify the checksum of the installer.
3. Run the readiness check and keep **lab_readiness_report.csv**. Verdict NOT READY: write to the instructor by **MIO** (Messagerie interne Omnivox, the internal messaging of the College) before Lab 1.
4. Watch the five videos of the lecture page: about 32 minutes together.
5. Create your free account at the **BAnQ** (Bibliothèque et Archives nationales du Québec).
6. Review the flash cards and the questions of the three lessons.





Links: official sources and further reading (1/3)

Cyber.SoHo, the College and the library

[Cyber.SoHo Legal Notice, Terms of Use and Acceptable Use Policy](#)

[Dawson College](#)

[Bibliothèque et Archives nationales du Québec](#)

Canadian public bodies and laws

[Canadian Centre for Cyber Security](#)

[Glossary of the Canadian Centre for Cyber Security](#)

[National Cyber Threat Assessment 2025-2026](#)

[Baseline cyber security controls for small and medium organizations](#)

[Get Cyber Safe](#)

[Statistics Canada, The Daily, October 21, 2024](#)

[Statistics Canada, The Daily, October 18, 2022](#)

[Canadian Anti-Fraud Centre](#)

[Commission d'accès à l'information du Québec](#)

[LégisQuébec: Act respecting the protection of personal information in the private sector](#)

[Office of the Privacy Commissioner of Canada](#)





Links: official sources and further reading (2/3)

[Justice Laws Website: Criminal Code](#)

Standards and frameworks

[Glossary of the National Institute of Standards and Technology: information security](#)

[Federal Information Processing Standards Publication 199](#)

[Cybersecurity Framework of the National Institute of Standards and Technology](#)

[International Organization for Standardization: the 27000 family of standards](#)

The lab environment

[VirtualBox: official site](#)

[VirtualBox: official download page](#)

[VirtualBox: user manual](#)

[VirtualBox: Personal Use and Evaluation License](#)

[VMware Workstation Pro and Fusion Pro: product page](#)

[Broadcom: Download and license VMware desktop hypervisor products](#)

[Broadcom support portal](#)

[Microsoft: Enable virtualization on Windows](#)

[Amazon Web Services Academy](#)





Links: official sources and further reading (3/3)

[Amazon Web Services Academy portal](#)

[Amazon Web Services: shared responsibility model](#)

[CyberQuébec](#)

[Ubuntu: official download page](#)

[Python: official site](#)

[Microsoft: PowerShell documentation](#)

Videos recommended on this page

[Simplilearn: What Is Cyber Security, How It Works? Cyber Security in 7 Minutes](#)

[Professor Messer: The Confidentiality, Integrity, Availability Triad](#)

[International Business Machines Technology: Cybersecurity Architecture, Fundamentals of Confidentiality, Integrity, and Availability](#)

[International Business Machines Technology: Virtualization Explained](#)

[GEEKrar: How to Install VirtualBox on Windows 11](#)

Tools to check a link or a file

[VirusTotal: check an address](#)

[VirusTotal: check a file](#)

[Google Safe Browsing: site status](#)



Questions?

Next class: Friday, September 18, 6:00 p.m., online.
Types of cyberattacks; prevention, detection and response.



© 2026 Cyber.SoHo — Educational Hub. All rights reserved. Educational content only.

Use of this presentation is subject to the [Cyber.SoHo Legal Notice, Terms of Use and Acceptable Use Policy](#).

No affiliation with any vendor, platform or institution. Product names belong to their owners. Fictional organization and data.

